

Top Tech Trends to Watch Out For in 2018

The technologies that will dominate the tech world in the new year are based on open source software.

At the start of a brand new year, I looked into the crystal ball to figure out the areas that no technologist can afford to ignore. Here is my rundown on some of the top trends that will define 2018.

Automation and artificial intelligence

Two of the most talked about trends are increasingly utilising open source. Companies including Google, Amazon and Microsoft have released the code for Open Network Automation Platform (ONAP) software frameworks that are designed to help developers build powerful AI applications.

In fact, Gartner says that artificial intelligence is going to widen its net to include data preparation, integration, algorithm selection, training methodology selection, and model creation. I can point out so many examples right now such as chatbots, autonomous vehicles and drones, video games as well as other real-life scenarios such as design, training and visualisation processes.

Open source containers are no longer orphans now

DevOps ecosystems are now seeing the widespread adoption of containers like Docker for open source e-commerce development.

Containers are one of the hottest tickets in open source technology. You can imagine them as a lightweight packing of application software that has all its dependencies bundled for easy portability. This removes a lot of hassles for enterprises as they cut down on costs and time.

According to 451 Research, the market is expected to grow by more than 250 per cent between 2016 and

2020. Microsoft recently contributed to the mix by launching its Virtual Kubelet connector for Azure, streamlining the whole container management process.

Blockchain finds its footing

As Bitcoin is likely to hit the US\$ 20,000 mark, we're all in awe of the blockchain technology behind all the crypto currencies. Other industries are expected to follow suit, such as supply chain, healthcare, government services, etc.

The fact that it's not controlled by any single authority and has no single point of failure makes it a very robust, transparent, and incorruptible technology. Russia has also become one of the first countries to embrace the technology by piloting its banking industry's first ever payment transaction. Sberbank, Russia's biggest bank by assets, has executed a realtime money transfer over an IBM-built blockchain based on the Hyperledger open source collaborative project.

One more case in point is a consortium comprising more than a dozen food companies and retailers — including Walmart, Nestle and Tyson Foods—dedicated to using blockchain technology to gather better information on the origin and state of food.

IoT-related open source tools/libraries

IoT has already made its presence felt. Various open source tools are available now that are a perfect match to further

the IoT challenges such as Arduino, Home Assistant, Zetta, Device Hive, and ThingSpeak. Open source has already served as the foundation for IoT's growth till now and will continue to do so.

OpenStack to gain more acceptance

OpenStack has enjoyed tremendous success since the beginning, with its exciting and creative ways to utilise the cloud. But it lags behind when it comes to adoption, partly due to its complex structure and dependence on virtualisation, servers and extensive networking resources.

But new fixes are in the works as several big software development and hosting companies work overtime to resolve the underlying challenges. In fact, OpenStack has now expanded its scope to include containers with the recent launch of the Kata containers project.

Open source is evolving at a great pace, which presents tremendous opportunities for enterprises to grow bigger and better. Today, the cloud also shares a close bond with the open source apps, with the services of various big cloud companies like AWS, Google Cloud, and Microsoft Azure being quite open source-friendly.

I can think of no better way to say this, but it is poised to be the driver behind various innovations. I'd love to hear your thoughts on other trends that will dominate 2018. Do drop me a line (dinesh@railsfactory.com). 

By: Dinesh Kumar

The author is the CEO of Sedin Technologies and the co-founder of RailsFactory. He is a passionate proponent of open source and keenly observes the trends in this space. In this new column, he digs into his experience of servicing over 200 major global clients across the USA, UK, Australia, Canada and India.